

Fold Six Napkins Decoratively SOP

One episode folds six napkins and stages them as one stack on a plate. Napkins come off the supply stack one at a time, are folded on the mat, and are laid on the plate in a fixed alternating order: triangle, fan, triangle, fan, triangle, fan. The supply spot is empty and the mat is bare when the episode ends.

Every napkin starts the same way, with one triangle fold. A fan napkin then gets three pleats folded across that triangle, each one swinging the opposite way from the one before it, and is folded in half at the middle so the pleats nest and the napkin opens as a fan. Nothing else changes from napkin to napkin.

Every fold is the same three moves: the left gripper pins, the right gripper carries the corner or the edge over and lays it down, and the right gripper draws the crease. The gather is the one fold that skips the crease, because a pressed spine holds the fan shut. The left arm never folds and the right arm never pins.

Nothing is turned in the air, nothing is slid across the mat except a crease stroke, and no gripper opens anywhere but the mat or the plate.

Setup

Complete both checklists before starting an episode.

Hardware checklist

1. Cameras are on and recording.
2. The environment camera shows the whole tabletop: the supply stack at the far side, the fold mat in the middle, and the serving plate on the right.
3. Both arms are at home with grippers open.
4. The tabletop is clear of anything but the zones listed below.
5. Nothing obstructs the paths between the supply stack, the mat, and the plate.

Materials checklist

1. Six matching square cloth napkins, pressed flat, dry, and free of stains and tears.
2. The six napkins sit in one squared stack on the supply spot, each lying as a diamond with one corner pointing at the front edge. No napkin overhangs the one below it.
3. The napkins separate cleanly. A napkin clinging to the one under it is replaced before the episode.
4. The fold mat lies flat in the middle of the tabletop and is bare. Its surface holds cloth still under a crease stroke rather than letting it skate.

5. The serving plate sits on the right, empty, right side up, and wide enough that a folded triangle rests on it without overhanging the rim.

Workspace layout

- **Supply spot:** the squared stack of six napkins at the far side of the tabletop, centered between the two arms so both grippers reach it without leaning across. It is the only place a napkin is picked up from.
- **Fold mat:** flat in the middle of the tabletop, reachable by both arms. Every fold and every crease happens here. It holds one napkin at a time and is bare at the start and at the end.
- **Serving plate:** on the tabletop against the right edge, within reach of the right arm. The finished stack builds here and nothing else is ever set on it.

Vocabulary

- **Arm assignment:** the left arm pins the cloth and helps lift a napkin off the supply stack. The right arm makes every fold, every crease, and every placement on the plate. Neither arm takes the other's job.
- **Diamond:** a napkin lying flat and open on the mat with one corner pointing at the front edge.
- **Napkin corners:** near, far, left, and right, named by where they point while the napkin lies as a diamond.
- **Pin:** a closed gripper pressed straight down on the cloth to hold it still. A pin never pinches the cloth and never travels.
- **Fold:** the right gripper pinches one corner or the middle of one edge, lifts it clear of the cloth below, carries it to its target, lowers it, and releases.
- **Crease:** the closed right gripper pressed down on a new fold edge and drawn along it in one continuous stroke, end to end.
- **Fold edge:** the closed folded side a fold creates, as opposed to the loose open edges.
- **Triangle napkin:** one triangle fold, creased, and nothing else. Napkins 1, 3, and 5.
- **Fan napkin:** the triangle fold, plus three creased pleats, plus the gather. It holds an open fan with no gripper on it. Napkins 2, 4, and 6.
- **Working edge:** the long fold edge of the triangle, the one facing the front edge when Step 2 ends. The right gripper takes hold of this same edge for all three pleats and swings it back and forth across the face.
- **Pleat:** one swing of the working edge. The right gripper pinches it, lifts it clear, carries it the opposite way from the last swing, and lays it down, which lays in a fresh crease.
- **Band:** the strip of cloth between one crease and the next. The three pleats leave bands that look about the same width as each other.
- **Swing:** the travel of one pleat, away from the front edge or back toward it. Every swing covers about the same distance, so the bands come out even.

- **Gather:** the last fold on a fan napkin. The pleated band is folded in half across its middle so the pleats nest into each other and the napkin springs open as a fan.
- **Gather point:** the folded middle the gather creates, where every pleat is held together. It is the only place a fan napkin is picked up.
- **Settled fold:** the napkin stays put for 2 seconds after the gripper lifts clear, with no layer springing open.
- **Squared on the stack:** centered on the napkin below it, long fold edge toward the front edge, point aimed away.
- **Release point:** the fold mat and the plate. A gripper opens nowhere else.

Steps

Steps 1 to 4 are one napkin's full cycle. Step 3 is done only on a fan napkin.

Step 1: Move a napkin to the fold mat

Goal: the top supply napkin lies flat and open on the mat as a diamond.

- The **left gripper** pinches the top napkin's **left corner** and the **right gripper** pinches its **right corner**. Take the top napkin only.
- Both grippers lift it straight up clear of the stack, carry it level toward the front edge, and lower it flat onto the middle of the mat.
- Release both grippers once the napkin is resting, then lift both clear.

Check: the napkin lies flat as a diamond with one corner pointing at the front edge, no corner folded under, and nothing hanging off a mat edge. If it lands crumpled or turned, lift it by the same two corners and lay it down again. If a second napkin comes up with it, lay both on the mat, return the extra one to the supply stack, and lay the remaining napkin out again.

Step 2: Make the triangle fold

Goal: a creased triangle with its long fold edge toward the front edge and its point aimed away.

- The **left gripper**, closed, presses down on the **left corner** and holds it there for the whole step.
- The **right gripper** pinches the **near corner**, lifts it clear of the cloth, carries it over the napkin, and lowers it onto the **far corner**. Release once it is resting.
- The **right gripper**, closed, presses down on the new fold edge at the **right corner** and draws along that edge in one stroke until it reaches the left gripper.
- Both grippers lift clear.

Lift the corner clear before carrying it. A corner dragged across the mat pulls the whole napkin with it and lands the fold off line.

Check: the napkin is a triangle, its long fold edge faces the front edge, the moved corner sits on the far corner rather than off to one side, the fold edge is creased flat along its whole length, and no layer has sprung open. If the corners are badly out of line, open the napkin back to a diamond and fold it again.

Step 3: Pleat the napkin and gather the fan

Done only on a **fan napkin**, which is napkin 2, 4, or 6. A triangle napkin goes straight from Step 2 to Step 4.

Goal: three creased pleats, folded in half at the middle, so the napkin holds an open fan on its own with nothing gripping it.

The step runs in two parts. Pleat first, then gather.

Pleat the napkin

All three pleats take hold of the same **working edge** and swing it the opposite way from the swing before. The distance never changes. Away from the front edge, back toward it, away again.

- The **left gripper**, closed, presses down on the triangle just short of its **far corner** and holds it there for all three pleats.
- **First pleat:** the **right gripper** pinches the middle of the **working edge**, lifts it clear, carries it away from the front edge, and lowers it partway up the face, well short of the far corner. Release once it is resting.
- The **right gripper**, closed, presses on the new crease at its left end and draws across to its right end in one stroke.
- **Second pleat:** the **right gripper** pinches the **working edge** again, lifts it clear, carries it back toward the front edge, and lowers it after about the same distance the first pleat travelled. Crease the new edge the same way.
- **Third pleat:** swing the **working edge** away from the front edge again, the same distance, and crease it.
- Both grippers lift clear.

Every pleat goes the opposite way from the last one. Two swings in a row going the same way roll the napkin up into a flat band and the zigzag is gone.

Keep the swings the same length. A short swing after a long one leaves one band much thinner than the rest and the fan looks lopsided.

Check: three creases run across the face, they sit parallel to each other, the bands between them look about the same width, the napkin holds a zigzag instead of lying flat, and every crease is drawn end to end. If a pleat went the wrong way or landed crooked, lift that pleat back up, swing it again, and crease it.

Gather the fan

The pleated napkin is now a wide band that runs left to right. Folding it in half across the middle nests the pleats into each other and the napkin springs open as a fan. This is what makes the fan hold its shape once both grippers are off it.

- The **left gripper**, closed, presses down on the **middle** of the pleated band and holds it there. This spot is the fold line and it stays put for the whole gather.
- The **right gripper** pinches the **right end** of the pleated band, lifts it clear of the mat, and carries it left over the top of the left gripper.
- Lower the right end onto the **left end** of the band so the two halves sit on each other and the pleats nest. Release once it is resting.
- Both grippers lift clear. The napkin opens into a fan and stays open.

The left gripper pins the middle and nothing else. Pinning off to one side puts the fold line off the middle, so one half of the fan is longer than the other and the pleats do not nest.

Do not crease this fold. A pressed spine locks the pleats shut and the fan will not open.

Check: the napkin holds an open fan with no gripper on it, the two halves are about the same length, the pleats sit nested rather than crossed, and the fold sits at the middle. If the fan collapses flat or the halves are plainly uneven, open the fold back out, square the band up, and fold it in half again.

Step 4: Stage the napkin on the plate

Goal: the finished napkin lies squared on the plate stack.

The two kinds of napkin are picked up in different places. A triangle goes by its edge. A fan goes by its middle, because that is the one spot where every pleat is held together.

- On a **triangle napkin**, the **right gripper** pinches the middle of the **near fold edge**, the folded edge closest to the front edge.
- On a **fan napkin**, the **right gripper** pinches the **gather point**, the folded middle where the pleats meet. Never lift a fan by a pleat or by an open end.
- Lift the napkin level off the mat and carry it to the plate.
- Lower it onto the plate, centered on the napkin below it. A triangle lands with its long fold edge toward the front edge and its point aimed away. A fan lands with its gather point toward the front edge and its open side aimed away. Release once it is resting.
- Lift the gripper clear. The stack settles under its own weight and nothing presses it down.

Carry the napkin level. Tilting it up on an open edge lets the layers slide apart in transit. A fan carried level by the gather point keeps its pleats nested the whole way.

Check: the napkin lies centered on the stack, no layer sprung open, and nothing overhanging the rim. A triangle has its fold edge toward the front edge. A fan still reads as a fan after it lands,

with its pleats spread and its gather point toward the front edge. If it lands skewed or off the stack, lift it by the same spot and lay it down again. If a layer sprang open or a fan collapsed, carry it back to the mat, fold it again from Step 2, and bring it back.

Step 5: Fold the remaining napkins

Goal: all six napkins are folded and stacked in the alternating order.

- Work the supply one napkin at a time, always taking the top one, and run Steps 1 to 4 on each.
- The pattern alternates by turn and never changes: napkin 1 triangle, napkin 2 fan, napkin 3 triangle, napkin 4 fan, napkin 5 triangle, napkin 6 fan.
- Finish a napkin completely and get it onto the plate before fetching the next one. The mat holds one napkin at a time.

Check: the plate holds six napkins, the supply spot is empty, and the mat is bare. Counting from the bottom, the stack runs triangle, fan, triangle, fan, triangle, fan.

Step 6: End the episode

- Confirm the plate holds six napkins in the alternating order, the mat is bare, and the supply spot is empty.
- Return both arms home, then stop recording.

After the episode: reset the workspace

This reset is not recorded.

1. Lift the six napkins off the plate and open each one out flat.
2. Press every napkin flat again so no fold line from the last episode is left in the cloth. A napkin that still shows an old crease is not restacked.
3. Restack the six pressed napkins squarely on the supply spot, each lying as a diamond with one corner toward the front edge. Vary the order the napkins go back in rather than rebuilding the same stack every time.
4. Replace any napkin that is torn, stained, damp, or clinging to its neighbor.
5. Wipe the plate, dry it, and set it back empty against the right edge.
6. Smooth the fold mat flat in the middle of the tabletop and leave it bare.
7. Wipe the tabletop.
8. Run both Setup checklists again.

SOP violations

These are actions that break the SOP and are reviewed side by side in the review tool.

How to record a violation in review

For each violation, record the **start timestamp**, **violation name**, and **SOP rule broken**. The visible cue is what the reviewer sees. The coaching note is for retraining and is not an annotation label.

Episode handling

Tag every violation with its timestamp and name. Keep the episode with the violation tag. Do not delete it just because a rule was broken.

Violations

Violation: Wrong pattern for the napkin's turn

- **Visible cue:** pleats are added to napkin 1, 3, or 5, or skipped on napkin 2, 4, or 6, so the finished stack does not alternate triangle, fan, triangle, fan, triangle, fan.
- **SOP rule broken:** Step 5, the pattern alternates by turn and never changes.
- **Coaching note:** name the pattern before you take the napkin. Odd turns are triangles.

Violation: Napkin taken from anywhere but the top of the stack

- **Visible cue:** a napkin is pulled from under another, slid out sideways, or lifted from the middle of the supply stack.
- **SOP rule broken:** Step 1 and Step 5 (always take the top napkin).
- **Coaching note:** top of the stack, every turn.

Violation: More than one napkin moved at once

- **Visible cue:** two napkins come off the supply together and both go onto the mat or the plate, or a second napkin is fetched while one is still on the mat.
- **SOP rule broken:** Step 1 and Step 5 (take the top napkin only; the mat holds one napkin at a time and a napkin is plated before the next is fetched).
- **Coaching note:** one napkin on the mat, start to finish.

Violation: Fold made without a pin

- **Visible cue:** the left gripper is off the cloth while the right gripper folds or creases, or it lifts away before the crease stroke is finished, and the napkin shifts under the motion.
- **SOP rule broken:** Steps 2 and 3, the left gripper presses down and holds for the whole step.
- **Coaching note:** pin first, let go last.

Violation: Arms swapped roles

- **Visible cue:** the left gripper makes a fold, draws a crease, or places a napkin on the plate, or the right gripper is used as the pin. Both grippers lifting a napkin off the supply stack in

Step 1 is not this violation.

- **SOP rule broken:** Steps 1 to 4, the left arm pins and helps lift off the supply, the right arm folds, creases, and plates.
- **Coaching note:** left holds, right works.

Violation: Fold made by dragging

- **Visible cue:** the corner or the fold edge is pushed, shoved, or slid across the mat into place instead of being pinched, lifted clear of the cloth, and lowered onto its target.
- **SOP rule broken:** Steps 2 and 3, lift the napkin clear before carrying it.
- **Coaching note:** pick it up and put it down. Nothing gets pushed.

Violation: Crease skipped or dabbed

- **Visible cue:** the gripper taps the fold edge in spots, presses once in the middle, or moves on to the next fold with no stroke drawn along the edge at all.
- **SOP rule broken:** Steps 2 and 3, one continuous stroke end to end along the fold edge.
- **Coaching note:** one stroke, corner to corner, without lifting off.

Violation: Pleat swung the wrong way

- **Visible cue:** two pleats in a row go the same direction, so the napkin rolls up into a flat band and no zigzag is left on the face.
- **SOP rule broken:** Step 3, every pleat goes the opposite way from the last one.
- **Coaching note:** away, back, away. Say it as you swing.

Violation: Pleats came out uneven

- **Visible cue:** the three bands are plainly different widths, or a crease runs at an angle to the ones next to it, and the episode moves on.
- **SOP rule broken:** Step 3, keep the swings the same length and the creases parallel.
- **Coaching note:** match the last swing before you put the edge down.

Violation: Gather skipped or done wrong

- **Visible cue:** a fan napkin goes to the plate as a flat pleated band with the halves never folded onto each other, or the left gripper pins away from the middle so one half ends up plainly longer than the other, or the right gripper draws a crease along the gather fold and the fan stays shut.
- **SOP rule broken:** Step 3, fold the pleated band in half at the middle so the pleats nest, pin the middle and nothing else, and do not crease this fold.
- **Coaching note:** the pleats are not the fan, the gather is. Middle for the pin, no stroke on the spine.

Violation: Napkin turned or regripped in the air

- **Visible cue:** a napkin's orientation changes while it is in a gripper, between the mat and the plate or during a fold, or the gripper shifts its hold along the cloth without setting the napkin down first.
- **SOP rule broken:** Step 4, carry it level and lower it as it is, and take a fresh hold only after the napkin is resting.
- **Coaching note:** put it down before you turn it or take a new hold.

Violation: Fan lifted by the wrong spot

- **Visible cue:** a fan napkin is picked up off the mat by a pleat, by an open end, or by an edge instead of by the gather point, and the pleats pull apart as it rises.
- **SOP rule broken:** Step 4, a fan is pinched at the gather point and never lifted by a pleat or an open end.
- **Coaching note:** fans travel by the middle. Only triangles go by the edge.

Violation: Released too high or over the wrong place

- **Visible cue:** a gripper opens above the mat or above the plate stack and the napkin falls the rest of the way, or a gripper opens over the supply stack, over bare tabletop, or over open space.
- **SOP rule broken:** Steps 1 to 4, the gripper opens over the fold mat and the plate and nowhere else, and only once the napkin is resting.
- **Coaching note:** all the way down, and know where it lands before you open.

Violation: Napkin shoved, patted, or pressed down

- **Visible cue:** the gripper nudges, pats, brushes, or straightens a napkin on the mat or on the plate after releasing it, or bears down on the plated stack to flatten it, square it up, or hold a layer closed.
- **SOP rule broken:** Steps 1 to 4 (correct a bad placement by lifting the napkin clear and setting it down again; the stack settles under its own weight and nothing presses it down).
- **Coaching note:** lift it and lay it again, or leave it alone. The weight of the napkin is the only press it gets.

Violation: Napkin plated out of square

- **Visible cue:** a napkin is left on the stack skewed, off center, overhanging the rim, laid with its fold edge away from the front edge, or with a layer hanging open, and the episode moves on.
- **SOP rule broken:** Step 4, centered on the napkin below, fold edge toward the front edge, point aimed away.
- **Coaching note:** check the landing before you go back for the next one.

Violation: Napkin dropped, collided, or knocked something over

- **Visible cue:** a napkin falls off the mat or the plate, the plate or the supply stack is struck and shifts, the stack is toppled, or a folded napkin comes apart in transit.
- **SOP rule broken:** Steps 1 to 5 (keep each napkin controlled on a clear travel path and release only after a settled placement).
- **Coaching note:** check the path and the landing spot before you move.

Violation: Wrong episode ending

- **Visible cue:** the episode ends with a napkin still on the mat, fewer than six on the plate, the supply spot not empty, an arm away from home.
- **SOP rule broken:** Step 6 (confirm the six plated napkins, the bare mat, and the empty supply spot; return both arms home; then stop recording).
- **Coaching note:** confirm first. Homing is the last thing the arms do.

Failures that are not violations

These failures are not caused by how the task was run. Log them as system issues, discard the episode, and never use them for coaching.

- **Recording stopped or paused during the episode** (recording system).
- **Camera dropped frames or lost its feed** (capture system).
- **Hardware fault on an arm:** gripper failure, drift, controller caused collision, or motor error.
- **Defective napkin:** found torn, stained, damp, or clinging to the one below so it cannot be lifted free. Replace it before the next episode.
- **Napkin will not hold a crease:** creased end to end and the fold still springs open every time. Press it before the next episode.
- **Napkin will not hold a fan:** pleated and gathered correctly and the cloth still flops shut because it is too limp to stand. Replace it with a napkin that has more body.
- **Fold mat shifts out of its spot** or bunches under the napkin during the episode.
- **Plate or supply stack shifts out of its zone** during the episode.

Annotation subtasks (from SOP)

1. Move one napkin from the supply stack to the fold mat
2. Pin a napkin on the mat
3. Fold a napkin corner over
4. Swing one pleat
5. Crease a fold edge
6. Gather a pleated napkin into a fan
7. Move one folded napkin from the mat to the plate
8. Return both arms home and end the episode